PROJECT PERFECT FIT

This session, Comp4050, will introduce students to a project by Thomax.

THE CLIENT

Thomax provides end-to-end logistics software for warehouse and transport management. It is a software company with offices in all major Australian cities, and internationally in the US, NZ, UK, Canada and Singapore, headquartered in Pymble.

Its two main products, .wms for warehouse management, and .tms for transport management, are highly configurable, and fully cloud-based SaaS solutions with no server to maintain for their client. They include a “connect to anything” gateway that uses open APIs to seamlessly connect with carriers, ERPs, accounting packages, robotics and more.

THE PROJECT

The Perfect Fit Project aims to develop an integrated solution that enables teams to consolidate items into optimally packed cartons, reducing wasted space and the number of boxes required. At its core, the project will deliver an optimisation engine that determines the optimal selection and arrangement of boxes given a set of items and available container sizes, and a mobile-first web interface to visualise packing layouts and inspect how items should be placed within each carton. A central portal provides access for both operational users and integration with the other sub-systems.

DIVISIONS AND TEAMS

Given the size of the class, we will have multiple divisions, each working independently on their own integrated solutions. Each division will consist of 3 (sub-)teams responsible for the following sub-systems:

- **FitSolver (Optimisation Engine)**
  The FitSolver team develops and maintains the core optimisation engine of Project Perfect Fit, serving as a replacement for existing packing libraries such as BoxPacker, with a focus on extensibility and advanced constraint handling. It is responsible for designing and implementing efficient algorithms for three-dimensional bin packing that support realistic requirements such as weight distribution, item orientation, and container constraints.
- **FitPortal (User Interface / Access Layer)**
  The FitPortal team plays a central role in Project Perfect Fit by acting as the primary point of interaction between users and the system’s optimisation capabilities. It serves both front-end users, such as operational staff who define orders, and back-end users who require access to the optimisation engine.
- **FitVisualizer (Visualisation)**
  The FitVisualizer team is responsible for presenting optimisation results in a clear, interactive, and insightful manner, using a mobile-first design approach to ensure accessibility across devices. It develops intuitive visual representations of packing solutions, enabling users to explore spatial arrangements.

Given the expected enrollment, we expect there to be up to 6 Divisions:

A) Atomic Fit
B) Bionic Fit
C) Cosmic Fit
D) Dynamic Fit
E) Epic Fit
F) Fantastic Fit

Expectations

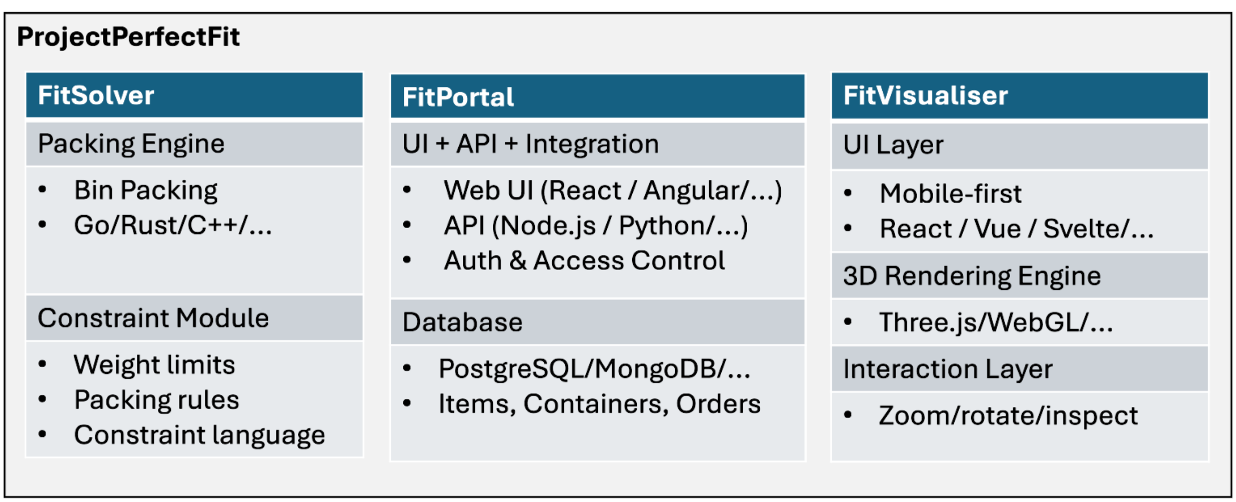
As a student, you will belong to a division, and in that division, in one sub-team. Your "team" will be your sub-team working on one of the sub-systems. We expect students to consistently contribute to their team. Each team is expected to be self-managed, setting its tasks and deliverables. However, this also requires coordination with other teams in the same division as well as with the client. Students will need to agree with the client on the technology stack, requirements, and milestones. They will also need to collaborate with other teams on interfaces, processes, handovers, and the division of tasks.

The teaching team is here to support this process. Their primary focus, however, is on running the unit and ensuring the achievement of its learning outcomes. This means they will provide guidance, deliver presentations on selected topics, and ensure that students collect the necessary artefacts for formative feedback, which then form the basis of the summative assessment. The teaching team will not manage the project.

Note that COMP4050 is not a PACE unit, and not an internship. Teams are expected to operate independently. Students are, for example, free to select their own technology stack. Neither the client nor the teaching team will manage the teams or facilitate interactions between them. Managing collaboration and coordination is one of the key learning outcomes of this unit.

Suggested Architecture

We suggest the following architecture and platforms.



Of course, your division can make your own decisions. Or, better, we expect you to make your own decisions.

SPRINTS

We will divide the session into 4 sprints, each approximately 3 weeks.

- Sprint 0: In this sprint you will be forming a team, pin down the requirements, decide on the technology stack, and just as important, agree with your teammates how the team functions, and with the other teams how you want to coordinate your work.
- Sprint 1: In this sprint, you will be working on producing an integrated MVP. This means you will have to agree with the other team in your submission how to integrate, and complete that integration at least once.
- Sprint 2: In this sprint, you will be implementing the main part of the functionality.
- Sprint 3: In this sprint, you will finalise the prototype and prepare deployment and delivery. No new feature should be implemented in this sprint.

LECTURES

About half of the week will start with a mini-lecture on selected topics:

- Agile refresher
- Working in teams
- DevOps
- Delivering effective technical presentations
- Reflecting on Skills: SFIA
- Protecting your IP

WEEKLY SESSIONS

The weekly sessions are held every Wednesday from 12:00 pm to 3:00 pm. In practice, these sessions serve as a focal point for working with your team and within your division. We expect all students to attend, not because of an attendance requirement, but because teams, especially newly formed teams, benefit greatly from frequent face-to-face communication.

If you are unable to attend, please notify your team before contacting any member of the teaching staff.

DELIVERABLES

Most sprints conclude with a retrospective and a sprint review, and we expect you to provide a record of these meetings. We also require individual logs, a mid-semester presentation, and a final presentation.

At the end of the unit, you will be required to deliver the product as a team (and division), along with an individual reflection. These documents and artefacts will all contribute to your assessment. A rubric available on iLearn explains how these artefacts will be assessed.

However, these artefacts are more than just assessment items. Treat each deliverable as an opportunity to capture a milestone, reflect on your progress, and identify improvements to both the product and the process.

SCHEDULE

	Week	Date	Topic	Activity	Deliverable
Sprint 0	1	29-Jul	**Welcome and Introduction**	Team formation	
	2	5-Aug	Agile refresher	**Meet the Client** Meet the peers	
	3	12-Aug		**Team Presentation** Sprint review, retrospective, sprint planning	Slides, logs
Sprint 1	4	19-Aug	DevOps and working in teams		
	5	26-Aug		***Client Q&A***	
	6	2-Sep			
	7	9-Sep		Sprint review and retrospective	Log, notes
Sprint 2	8	16-Sep		**Mid-term presentation**	Slides
	9	7-Oct	Delivering effective technical presentations	**Online Q&A**	
	10	14-Oct		Sprint review and retrospective	Logs, notes, peer feedback
Sprint 3	11	21-Oct	Reflecting on Skills: SFIA	**Client Q&A**	
	12	28-Oct	Protecting your IP		Repo
	13	4-Nov		**Final presentation and demo**	Slides, reflection, log